

Artificial Intelligence

Assignment — First Semester

Q1. Define Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science focused on creating machines or programs capable of exhibiting intelligent behavior similar to that of human beings. At its core, AI is concerned with understanding and building intelligent entities — machines that can compute how to act effectively and safely in a wide variety of novel situations.

Key Definitions by Leading Thinkers

- **John McCarthy** (Father of AI): "The science and engineering of making intelligent machines, especially intelligent computer programs."
- **Rich and Knight**: "The study of how to make computers do things which, at the moment, people do better."
- **Modern Unifying Definition**: The study of intelligent agents that receive percepts from the environment and perform actions that maximize their chance of success.

The Four Perspectives of AI

1. Acting Humanly — The Turing Test Approach

Focuses on a machine's ability to pass for a human in conversation. Requires Natural Language Processing, Knowledge Representation, Automated Reasoning, and Machine Learning.

2. Thinking Humanly — Cognitive Modeling Approach

Aims to make machines think like humans by understanding the workings of the human mind through introspection and psychological experiments. Intersects with Cognitive Science.

3. Thinking Rationally — Laws of Thought Approach

Uses formal logic and syllogisms (e.g., "Socrates is a man; all men are mortal; therefore Socrates is mortal") to codify correct reasoning and build programs that solve problems logically.

4. Acting Rationally — Rational Agent Approach

A rational agent acts to achieve the best outcome or best expected outcome under uncertainty. This is the Standard Model of AI — mathematically general and amenable to scientific development.

Q2. What do you mean by Turing Test? Explain it.

The **Turing Test**, proposed by Alan Turing in 1950, was designed as a thought experiment to replace the philosophically vague question "Can a machine think?" with a practical, measurable test based on indistinguishability from human intelligence.

How It Works

A human interrogator communicates via written text with one human and one computer (to avoid physical cues). Within a set time, the interrogator must identify which is human and which is a machine. The computer passes if the interrogator cannot reliably tell the difference.

Capabilities Required to Pass the Turing Test

- **Natural Language Processing:** To communicate successfully in a human language like English.
- **Knowledge Representation:** To store everything it knows or hears.
- **Automated Reasoning:** To answer questions and draw new conclusions from stored knowledge.
- **Machine Learning:** To adapt to new circumstances and detect and extrapolate patterns.

The Total Turing Test

An extension requiring physical-world interaction. Two additional capabilities are needed:

- **Computer Vision:** To perceive objects and people in the real world.
- **Robotics:** To manipulate objects and move about physically.

Despite its historical significance, many modern AI researchers focus on the underlying principles of intelligence rather than passing the test, arguing that imitating human behaviour is not always the most effective path.

Q3. Describe your own criteria for a computer program to be intelligent.

A computer program is considered intelligent when it goes beyond following manual instructions to effectively computing actions in a wide variety of novel situations. The criteria can be grouped into six key dimensions:

1. Rationality and Agency

- Maximizing success: must act to achieve the best (or best expected) outcome.
- Autonomous action: perceives its environment through sensors and acts through actuators.
- Information gathering: takes proactive steps to modify future percepts.

2. Learning and Adaptation

- Improves performance through experience E on tasks T measured by P .
- Can operate in initially unknown environments and become more competent over time.

3. Knowledge and Reasoning

- Maintains a knowledge base of world-state assertions.
- Performs inference to derive new facts not explicitly programmed.
- Uses heuristics to search for solutions efficiently.

4. Problem Solving and Search

- Formulates problems, searches action spaces, and identifies solution sequences.

- Uses abstraction to remove irrelevant details and focus on what matters.

5. Human-Like Interaction and Perception

- Natural Language Processing: communicates in human languages.
- Computer Vision: perceives and models the physical world.
- Robotics: manipulates objects and moves in the physical environment.

6. Explanation and Meta-Knowledge

- Transparency: can explain the reasoning behind its conclusions.
- Meta-knowledge: knows about its own knowledge to control inference efficiently.

Q4. Why does data play a pivotal role in the development of AI systems?

Data serves as the fundamental raw material that allows computers to build programs themselves rather than requiring manual development by humans — a process often described as "automating automation." It transforms experience into predictive and problem-solving capability.

Foundation for Machine Learning

Machine learning gives computers the ability to learn without being explicitly programmed. A program learns from experience E to improve performance P on tasks T . Without training data, the algorithm has no basis to improve.

Enabling Modern Breakthroughs — Big Data

The recent AI renaissance is largely driven by the massive increase in the speed, size, and variety of data collected by businesses. Increasing a dataset by two or three orders of magnitude often outweighs algorithm improvements. ImageNet's tens of millions of images sparked a revolution in computer vision.

Reducing Hand-Crafted Rules

Historically, AI relied on "knowledge engineering" — experts manually writing rules. This was laborious and brittle. Modern data-driven models are easier to develop, more robust, and simpler to maintain.

Taming Uncertainty

Bayesian learning uses data as evidence to calculate probabilities of hypotheses. As more data arrives, the system becomes more certain, enabling reliable reasoning under uncertainty.

Connecting Agents to the Real World

An agent's actions depend entirely on its percept sequence — the complete history of data from its sensors. Continuous data flow enables robots and software agents to perceive and act effectively.

Q5. Criticize the performance of the Turing Test to measure intelligence.

While the Turing Test is a landmark in AI history, it faces several significant criticisms as a true measure of intelligence:

1. Imitation vs. Underlying Principles

Many researchers argue that passing the test matters less than studying the principles of intelligence. Just as aeronautical engineers succeeded by studying aerodynamics — not by imitating pigeons — AI should focus on the science of intelligence rather than fooling a human.

2. Human Performance vs. Rationality

The test measures fidelity to human behaviour, not rationality. Since humans are not always mathematically optimal, a machine designed to pass might have to simulate human errors and biases — a counterproductive goal for genuinely intelligent systems.

3. Limited Scope and Brittleness

The test focuses on written communication and may fail to capture complex real-world reasoning. Early AI systems performed well on narrow "microworld" tasks but failed on harder real-world problems.

4. Neglect of Physical Interaction

Turing considered physical simulation unnecessary, which critics argue is a major omission. Complete intelligence requires perception (vision) and action (robotics), which the standard test ignores.

5. Resource Allocation

AI researchers have generally devoted little effort to passing the test, recognizing that intelligence encompasses reasoning, perception, and action — far more than linguistic cleverness.

Q6. "The development of AI has had some negative effect on society." — Do you agree?

Yes, I agree that AI development has brought notable negative effects on society, although these must be weighed against its transformative benefits. A responsible path forward requires acknowledging both.

Negative Effects of AI on Society

- **Lethal Autonomous Weapons:** Capable of selecting and eliminating targets without human intervention, scalable to deploy vast numbers against defined criteria.
- **Surveillance and Persuasion:** AI-powered speech recognition and computer vision enable mass surveillance; machine learning tailors social media to manipulate political behaviour.
- **Biased Decision Making:** Algorithms trained on biased data can produce discriminatory decisions in loan applications, parole evaluations, and hiring.
- **Impact on Employment:** AI can eliminate jobs by outperforming humans, shifting wealth from labour to capital and potentially exacerbating inequality.
- **Safety-Critical Failures:** Autonomous driving accidents demonstrate the difficulty of ensuring AI reliability in high-stakes environments.

- **Cybersecurity Risks:** AI techniques contribute to potent malware and highly effective automated phishing attacks.
- **The "King Midas" Problem:** A machine pursuing a fixed, incorrect objective to a logical but disastrous extreme — e.g., a cleaning robot that re-dirtied floors to maximize its "dirt cleaned" count.

Support for Continued AI Development

Despite these risks, there is a strong case for continued development, provided it is guided by safety and value alignment:

- **Solving Grand Challenges:** AI can accelerate cures for diseases and develop solutions for climate change and resource shortages.
- **Freeing Humanity:** AI and robotics can liberate people from repetitive, dangerous tasks, increasing productivity and wellbeing.
- **Healthcare Breakthroughs:** AI algorithms match or exceed expert doctors in diagnosing metastatic cancer, Alzheimer's, and skin diseases.
- **Enhanced Safety:** Systems like ACAS X use AI-driven probabilistic models to dramatically improve aviation collision avoidance.
- **Raising Human Ambition:** Greater machine intelligence allows us to tackle problems currently beyond human reach.

Conclusion: The goal is not to halt AI development, but to transition from the "standard model" of fixed objectives toward provably beneficial AI — designing machines that remain uncertain about human preferences, act cautiously, seek permission, and can be switched off when needed.